



Society of Petroleum Engineers

SPE-227424-MS

The Use of In-Well Flow Control Devices to Address Multiple Challenges of Carbon Dioxide Injection in C.C.S. Projects

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

The objective of this paper is to describe several challenges faced by Carbon Capture and Storage projects, how in-well flow control devices can be used to address the challenges, and the benefits and value of designing CO₂ injection wells with these capabilities.

Whether a CCS project targets injection in a deep aquifer or a depleted hydrocarbon reservoir, the challenges for CCS injection are: unbalanced injection rates into the leading to uneven growth of the CO₂ plume; back-flow of reservoir fluids into the wellbore or cross-flow between zones during well shut-in causing corrosion and formation of hydrates or scale; flashing of liquid or supercritical phase CO₂ to gas in the injection conduit when injecting into a pressure depleted reservoir causing extreme temperature variations that result in high thermal stress on casing and cement; and, loss of containment in the injection formation or compromise of well integrity.

Key to the formulation of solutions is understanding of the expected operating environment. Production scenarios considered are transient start-up, shut-down, and steady state injection, as well as extreme conditions, such as loss of pressure and flow containment at the wellhead. Injection flow and pressure requirements have been gathered, and a CO₂ injection simulator has established the environmental conditions. Operating parameters, functional specifications and qualification testing requirements have been defined for flow control devices for CCS well completions.

While depleted oil and gas fields can provide significant storage capacity for CCS, injection of carbon dioxide into pressure depleted hydrocarbon fields can be problematic due to the phase change of CO₂ that can cause flow instability and result in adverse physical phenomena, such as extreme cooling associated with the heat of vaporization. One solution is to use variable, autonomous or surface-controlled, flow control devices strategically placed in the injection tubing string to provide a stepwise pressure reduction and to maintain sufficient back pressure to prevent CO₂ flashing.

Distribution of CO₂ injection rates can be controlled with the use of appropriately sized passive ICDs, or by using autonomous outflow control devices. When the passive ICDs or autonomous OCDs are combined with a check-valve element, back-flow or cross-flow of reservoir fluids during injection shut-in can be prevented.

This paper continues the work of SPE-209705 and SPE-210651, proposing the use and performance modelling of flow control devices installed in the injection tubing to maintain carbon dioxide in a dense phase (liquid or super-critical) from the wellhead to reservoir interface, and to improve the efficiency of injection wells transporting CO₂ to the storage reservoirs.

Introduction

The relationship between climate change and greenhouse gases has been recognized by the scientific and meteorological communities for over 60 years. As the concentration of greenhouse gases (primarily carbon dioxide and methane) increase in the atmosphere, they trap solar radiation in the troposphere and stratosphere, leading to heating and increased temperatures of the atmosphere, oceans and land. This increase in temperatures has been associated with significant climate changes that affect global ecology and human endeavors such as agriculture, industry, and habitable land. Historical measurements of concentrations of carbon dioxide in the atmosphere, oceans, and polar and glacial ice show that the current concentrations are greater than they have been for the past 800,000 years, even after accounting for natural fluctuations. ([United States EPA – Climate Change Indicators Website 2025](#))

To address climate change and its negative impacts, the United Nations established the UN Framework Convention on Climate Change (UNFCCC), that was adopted at the 1992 Earth Summit and was one of the first international treaties on Climate Change. The Convention stipulated that parties should meet regularly to address climate change and established the foundation for future climate agreements. Under the Convention, annual meetings called Conference of Parties (COP) were held (Berlin, Geneva, Kyoto, etc.) establishing protocols and accords for monitoring and reducing CO₂ emissions, leading to the UN Climate Change Conference (COP21) at which world leaders reached a breakthrough agreement on 12 December 2015: the historic Paris Agreement. (www.un.org/en/climatechange/paris-agreement)

The Agreement includes commitments from all participating countries to reduce their emissions and work together to mitigate the impacts of climate change, with goals to hold global atmospheric temperature increase to below 2°C above pre-industrial levels and pursue efforts to limit it to 1.5°C. The Agreement is a legally binding international treaty, entered into force on 4 November 2016. As of January 1 2025, 195 Parties (194 States plus the European Union) have joined the Paris Agreement.

The Paris Agreement marks the beginning of a shift towards a net-zero emissions world, that is, "cutting carbon emissions to a small amount of residual emissions that can be absorbed and durably stored by nature and other carbon dioxide removal measures, leaving zero additions to the atmosphere." ([United Nations – Climate Action website 2025](#)) The greenhouse gas emissions need to be reduced by 45% by 2030 and reach net zero by 2050 to keep global warming to no more than 1.5°C as called for in the Paris Agreement.

Most of the reductions in CO₂ emissions will come from improvements in energy usage efficiency in transportation, commercial and residential (lighting, heating, air conditioning), electric power generation (from fossil fuels) and transmission, and industrial processes. Second, using renewable (solar, geothermal, wind, tide) energy sources to replace fossil fuel as an energy source has the next biggest impact on carbon emissions. Third, Carbon Capture and Sequestration (CCS) will address hard-to-decarbonize sectors, including fossil fuel electrical power generation, cement and steel production, Direct Air Capture (DAC), hydrogen generation from reforming natural gas, ethanol production, and other industrial sources.

The CCS value chain is comprised of three elements: 1) The capture of carbon dioxide at the source of emission; 2) the transportation of carbon dioxide from the source to the storage location, and 3) the permanent storage of carbon dioxide in a sub-terranean location. The oil and gas industry is positioned to contribute technology, methods, processes and management of the CCS value chain. From chemical, refinery, gas and LNG processing, to LNG, oil, and gas transport by pipeline, ship, rail and truck, to the drilling and construction of wells for injection and safe storage of carbon dioxide, including knowledge of geology, geomechanics, seismic analysis, and reservoir management and monitoring.

In 2024, carbon sequestration capacity amounted to approximately 41 million tonnes/year. DNV projects that the CCS capacity will quadruple by 2030 (270 MtCO₂/yr) and will increase to 1.3 billion tonnes/yr by the year 2050. DNV sees 2025 as a turning point for CCS – a point where the annual CCS capacity will accelerate significantly. (DNV - 2025). The technology building blocks for this growth are well established. The biggest barriers to the growth of CCS capacity is regulatory and fiscal policy uncertainty.

Thermodynamic Processes of CCS Injection

The thermodynamic processes of CCS injection are complex as they span the phase diagram for carbon dioxide, including liquid, vapor, and supercritical phases. The addition of small amounts of impurities (N₂, NO₂, CO, CH₄, C₂H₆, NH₃, etc.) adds complexity to the phase behavior. Carbon dioxide is usually captured at industrial or chemical plants in the vapor phase but is stored and transported in the dense liquid or supercritical phase. This increases the efficiency of temporary storage, transportation, and injectivity. Maintaining carbon dioxide in a single phase also prevents slugging and operability problems.

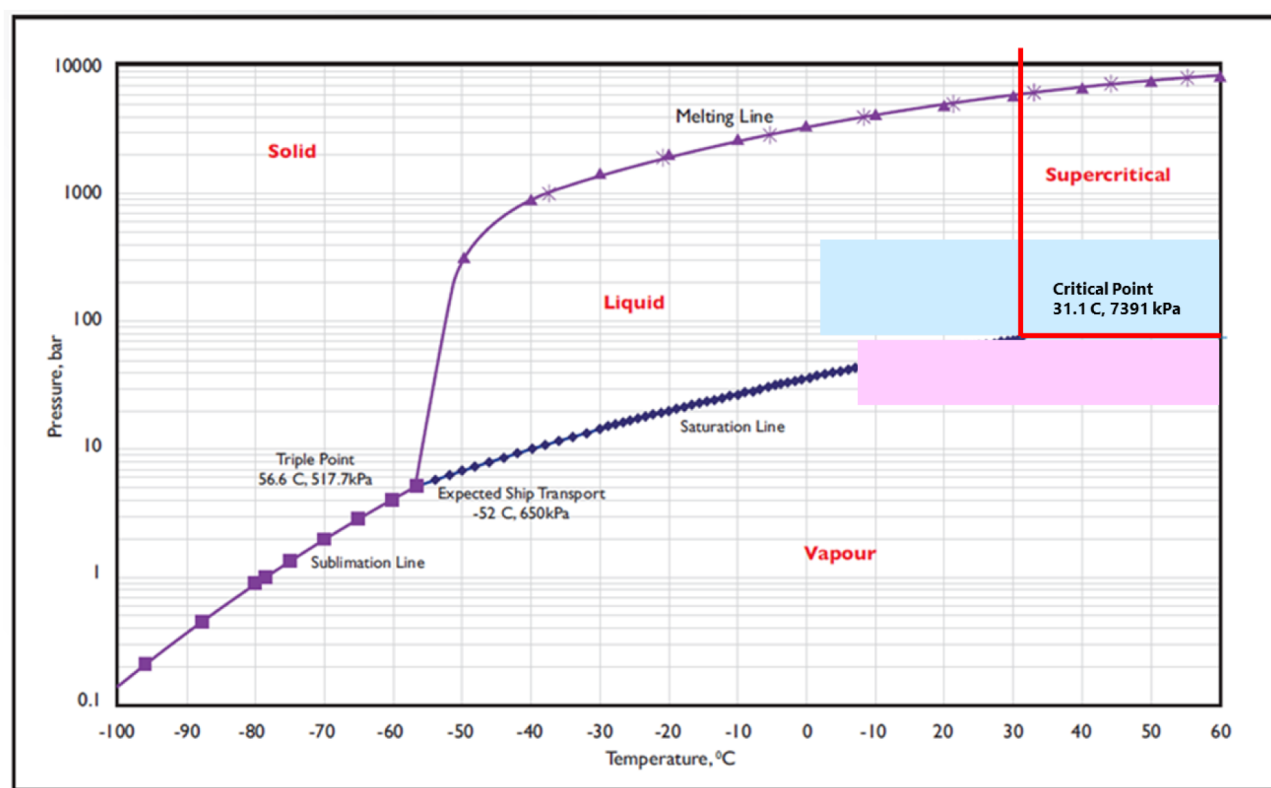


Figure 1—Phase diagram of CO₂: blue box represents typical CCS operating region for saline aquifer; pink box represents typical CCS operating region for depleted hydrocarbon reservoir. (SMITH, M. et al., 2012)

When dense phase carbon dioxide is delivered to the wellhead for injection into a depleted reservoir, if the reservoir pressure is insufficient to support a column of liquid or supercritical fluid, as the carbon dioxide flows from wellhead into the targeted formation, depressurization will occur, and the carbon dioxide will be forced to cross the saturation line into a vapor. Two important thermodynamic processes come into play with this transition: 1) Joule Thomson cooling and 2) the latent heat of vaporization. When either or both these processes are done with no heat exchanged with the environment – an adiabatic process – such as a rapid expansion through a throttling nozzle, the temperature of the carbon dioxide will decrease significantly.

Incorrectly, many practitioners of CCS projects often lump both processes together under the label of "Joule Thomson" (JT) cooling. This is incorrect. JT cooling (or heating) describes the temperature change

of a real gas or liquid when it is expanded (or compressed) in an adiabatic process. JT cooling does not include a change in phase of the fluid. In fact, the change in temperature due to JT cooling is much less than the cooling associated with the isenthalpic temperature change from the latent heat of vaporization when carbon dioxide expands from a dense phase to a vapor.

Note that if the reservoir pressure is sufficient to support a mixed column of gas and liquid CO₂, the vapor phase carbon dioxide may re-condense to a liquid phase on the way down the tubing, liberating the latent heat of condensation and reheating the fluid.

Heat transfer from the overburden and the reservoir is also an important thermodynamic process affecting CCS injection. As the carbon dioxide travels down the injection tubing, the amount of enthalpy it gains from heat transfer will depend on the velocity of the fluid, the temperature difference between the overburden and the fluid, and the heat capacity and thermal transmissibility of the overburden rock, and of the cement, casing, annular completion fluid and tubing that comprises the wellbore construction. For the same mass flow rate, carbon dioxide in the vapor phase is less dense than liquid or supercritical CO₂ and as a result, will travel at a higher velocity down the tubing and thus, will have a lower gain in enthalpy than the dense phase. (Figure 2) It is beneficial to keep the CO₂ in a dense phase during its traverse down the tubing to maximize its increase in enthalpy before reaching the reservoir.

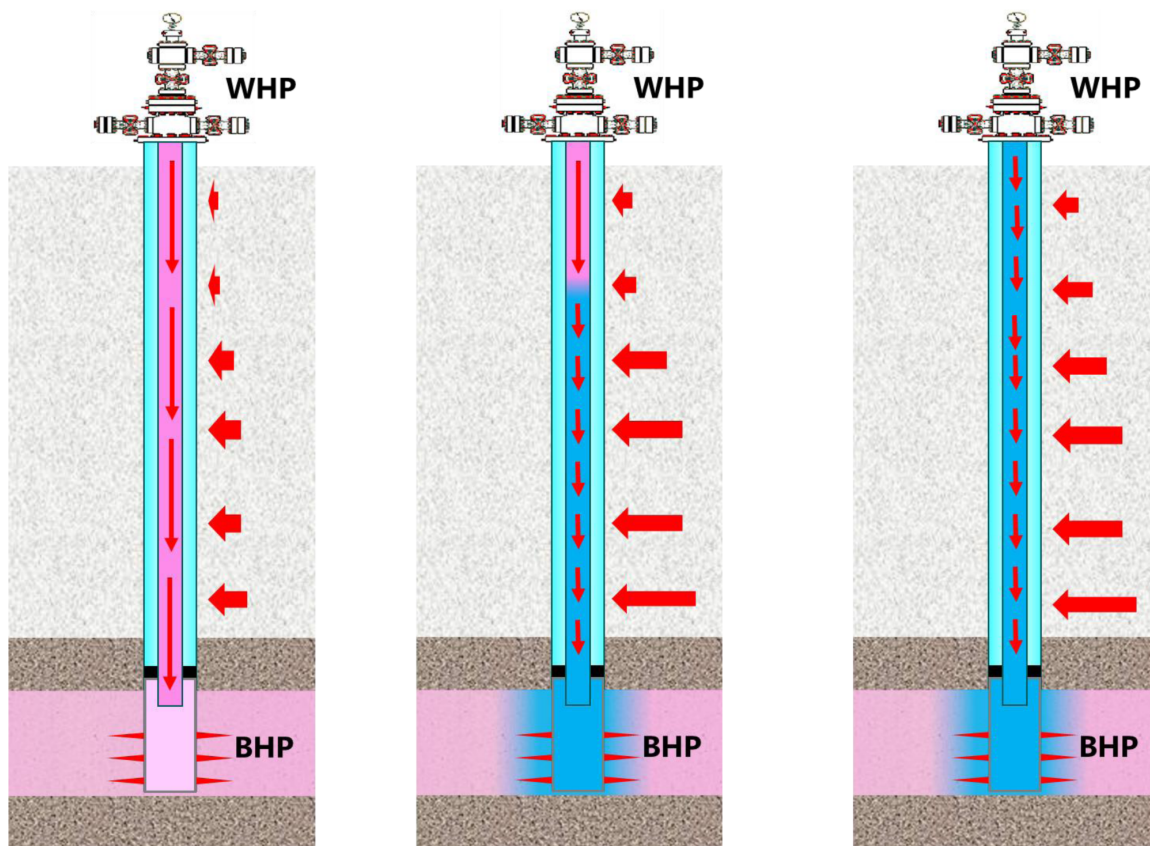


Figure 2—Heat transfer from the overburden for gas phase, mixed phase and dense phase CO₂ injection

CCS Reservoir Challenges

When dense phase carbon dioxide is injected into a low-pressure reservoir, the cooling due to the vaporization of the CO₂ and subsequent JT-effect can result in the possibility for ice or hydrate propagation in the pore space that can considerably impair injectivity (Hoteit, Fahs and Soltanian 2019, Hughes 2009, Maloney and Briceno 2009), freeze residual pore water (Loeve, Hofstee and Maas 2014), and fracture the

reservoir due to thermal stresses leading to increased transmissibility and injectivity (Jamaloei and Ashgari 2015).

If the carbon dioxide is injected into a multi-layered reservoir or one with substantial laminations, or into multiple separate reservoirs, the distribution of the carbon dioxide is likely to be unbalanced. This will result in unequal plume development, and may result in a limitation of the project storage capacity.

CCS Injection Well Challenges

The extreme temperature changes associated with the phase change from liquid CO₂ to vapor can be problematic for CCS injection wells. In an adiabatic process, the latent heat of vaporization must be extracted from the fluid itself, and as a consequence, the temperature of the fluid is reduced. Usually, this phase change occurs across the wellhead injection flow control valve (choke). The highly cooled vapor phase CO₂ then descends down the injection tubing in the well, gaining heat from the overburden via the cement, casing, and completion fluid. If the reservoir pressure is sufficient to support a mixed column of gas and liquid CO₂, the vapor phase carbon dioxide may re-condense to a liquid phase on the way down the tubing, liberating the latent heat of condensation and reheating the fluid.

Carbon sequestration projects can target injection in a deep aquifer or a depleted hydrocarbon reservoir, but several challenges for CCS injection are common:

1. unbalanced injection distribution into the reservoir resulting in uneven growth of the CO₂ plume;
2. back-flow of reservoir fluids into the wellbore or cross-flow between zones during well shut-in causing corrosion and formation of hydrates or scale, or back flow of formation solids that can fill the;
3. flashing of liquid or supercritical phase CO₂ to vapor in the injection tubing while injecting into a pressure depleted reservoir that causes extreme temperature variations resulting in high thermal stress on casing and cement; and,
4. loss of containment in the injection formation or compromise of well integrity.

Many of these challenges can be addressed by the use of flow control devices, including passive and active devices.

Passive and Active Flow Control Solutions for the Wellbore / Reservoir Interface

At the simplest end of flow control devices are the passive flow control devices, often called ICD's (Inflow Control Devices). ICD's come in a variety of flavors, including orifices, nozzles, tubes, helix and labyrinth or hybrid types (Figure 3). By far the simplest passive ICD's are the orifice or nozzle types, that generate an increasing pressure drop across the device as a function of increasing flow rate through the device. When deployed in sand screens (Figure 4) or mandrels and placed at the reservoir depth (across from the "sand face"), appropriate sizes and numbers of ICD's can balance the distribution of CO₂ injection amongst different layers of a reservoir or along long reservoir sections.



Figure 3—Types of Passive ICD's (flow control devices)

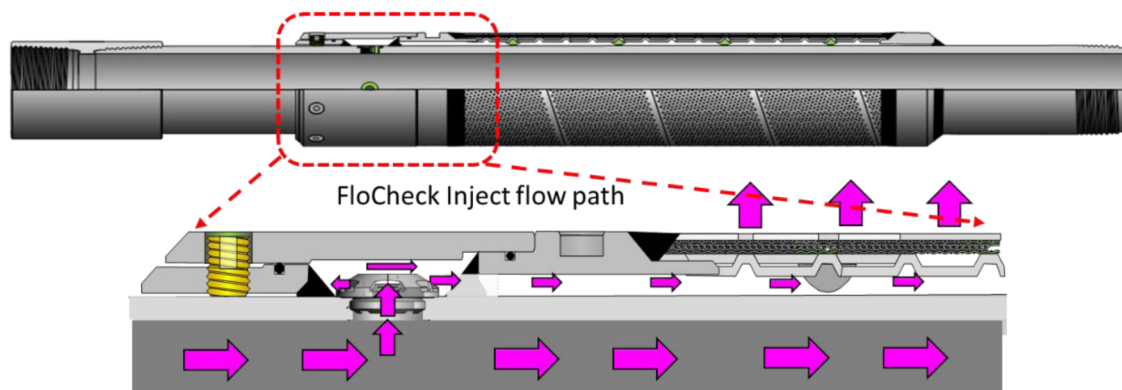


Figure 4—Deployment of a flow control device in a screen for injection

To be effective in balancing the flow distribution, the average pressure drop across the flow control device should be approximately equal to the average over-pressure on the reservoir (wellbore pressure – reservoir pressure). Too small a pressure drop across the ICD, and the flow distribution is ineffective. Too great a pressure drop across the ICD and the well requires more energy to achieve target rates and potentially can restrict the amount of CO₂ being injected. (Figure 5) Combined with swell packers to segment or compartmentalize the wellbore, this type of completion allows the best control of the distribution of CO₂ into multiple reservoir layers.

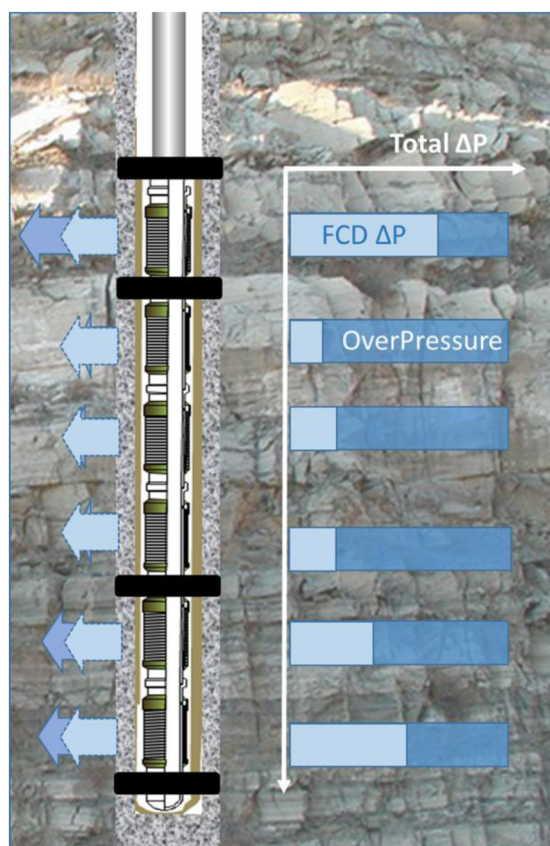


Figure 5—Integrated ICDs and screens with packers creating injection compartments to control injection distribution

Another passive flow control device that is useful in preventing back-flow and crossflow of reservoir fluids into the wellbore is the simple injection check valve. An example of this is shown in Figure 6. The valve allows flow when injecting and blocks flow in the reverse direction. The valve has a metal-to-metal

seal, and a pre-loaded spring to ensure positive shut-off. This device can be combined with an appropriately sized orifice type ICD for dual effectiveness – it can be used to control flow distribution and to prevent back-flow of reservoir fluids.

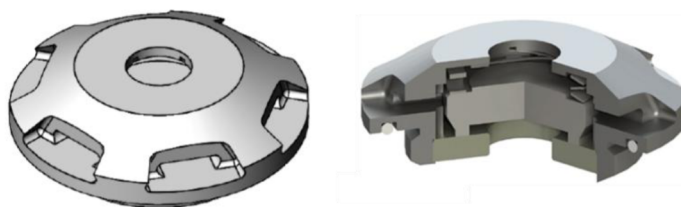


Figure 6—CO₂ Injection Check Valve

The flow performance of these devices with CO₂ can be calculated and established from flow loop testing. (Figure 7) This information can be used to establish the number of devices required for flow distribution and to maximize the total well injection capacity.

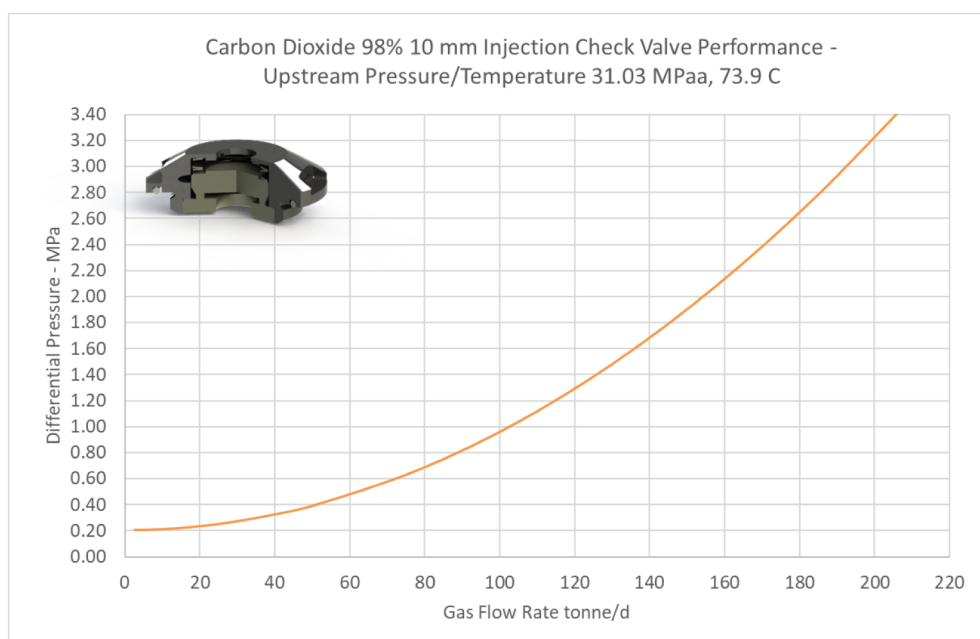


Figure 7—Injection Check Valve Flow Performance

Another deployment of the injection check valve can be seen in Figure 8. In this case multiple injection check valves are deployed on a mandrel and hung below a lock mandrel in a seating nipple located above the injection tubing packer. The number of injection check valves deployed are sufficient to handle the maximum target CO₂ injection rate. This simple deployment prevents the back flow of fluids into the wellbore, but does not prevent cross-flow between reservoir layers.

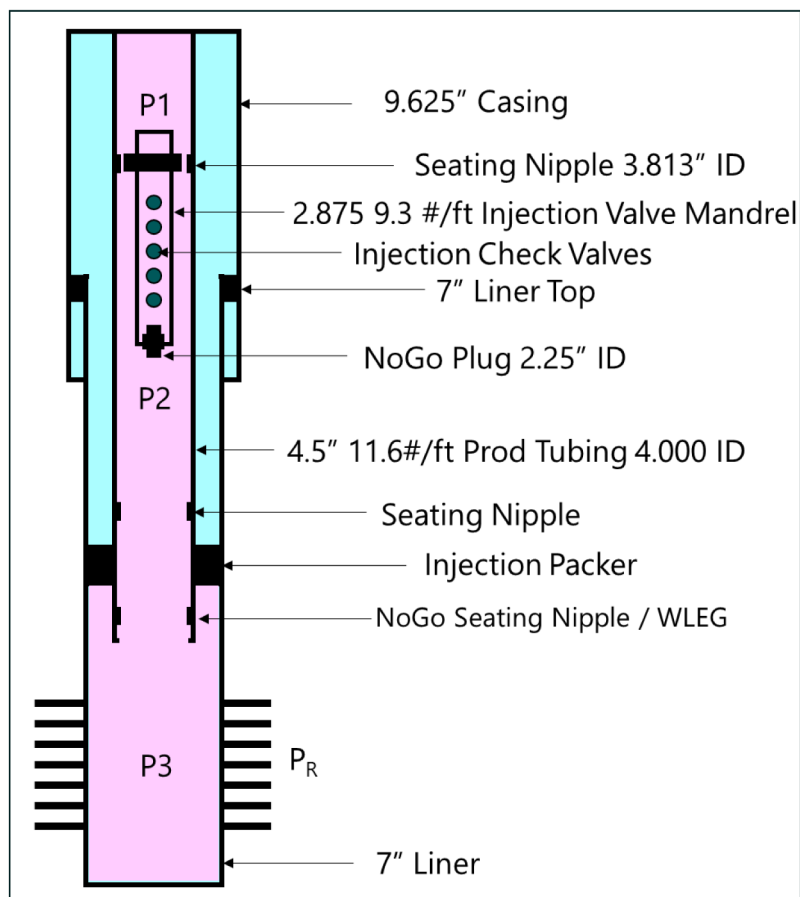


Figure 8—Wireline retrievable injection Check Valve deployment

Flow Control Solutions for the Upper Completion: Controlling the Phase Behavior of the CO₂

Carbon dioxide injection fluids should be kept in the liquid or supercritical phase during steady-state injection of CO₂ into a depleted reservoir to ensure a stable hydrostatic head, to maximize heat transfer from the overburden, to avoid potential hydrate formation and to prevent equipment damage associated with cooling by thermodynamic flashing and the Joule-Thomson effect. Methods of maintaining pressure include tubing friction or a reliance on sufficient back pressure from the reservoir due to high reservoir pressure or low injectivity.

Unstable injection rates can result from the phase-change behavior of the CO₂ and pressure gradient in the injection tubing. At the beginning of CO₂ injection in a depleted hydrocarbon reservoir, the low reservoir pressure results in a large pressure drop between the wellhead and the reservoir, requiring more back pressure in the tubing to keep the CO₂ in dense phase. Flashing will occur in the tubing downstream of the wellhead injection choke when the reservoir pressure is too low to support the liquid column of CO₂ in the tubing. The hydrostatic head of the CO₂ in the tubing will change significantly when flashing is allowed to occur downstream of the injection choke or in the injection tubing, resulting in lower downhole injection pressure, and a significant reduction in injection rate. If the CO₂ delivery rate is greater than the reservoir injection rate, the excess CO₂ delivered to the wellhead from the carbon capture source will accumulate in the space in the production tubing occupied by vapor phase CO₂. Pressure below the wellhead increases to accommodate the continual CO₂ supply, compressing and collapsing the CO₂ vapor back to liquid in the tubing, increasing the hydrostatic head and leading to a surge in CO₂ injection. Later in the well life, the

pressure in the reservoir will rise as the volume of CO₂ injected in the reservoir increases, and CO₂ flashing may not be significant. (Konopczynski and Pettitt – 2022)

To effectively control the pressure drop along the injection tubing and thus control the phase of the CO₂, variable, autonomous or surface-controlled, flow control devices strategically placed in the injection tubing string are recommended to provide stepwise pressure drops with depth. This flow control completion allows normal tubing sizes in the CCS applications, providing a greater range of injection rates without the need for multiple wells or to perform frequent, expensive workovers. (Figure 9)

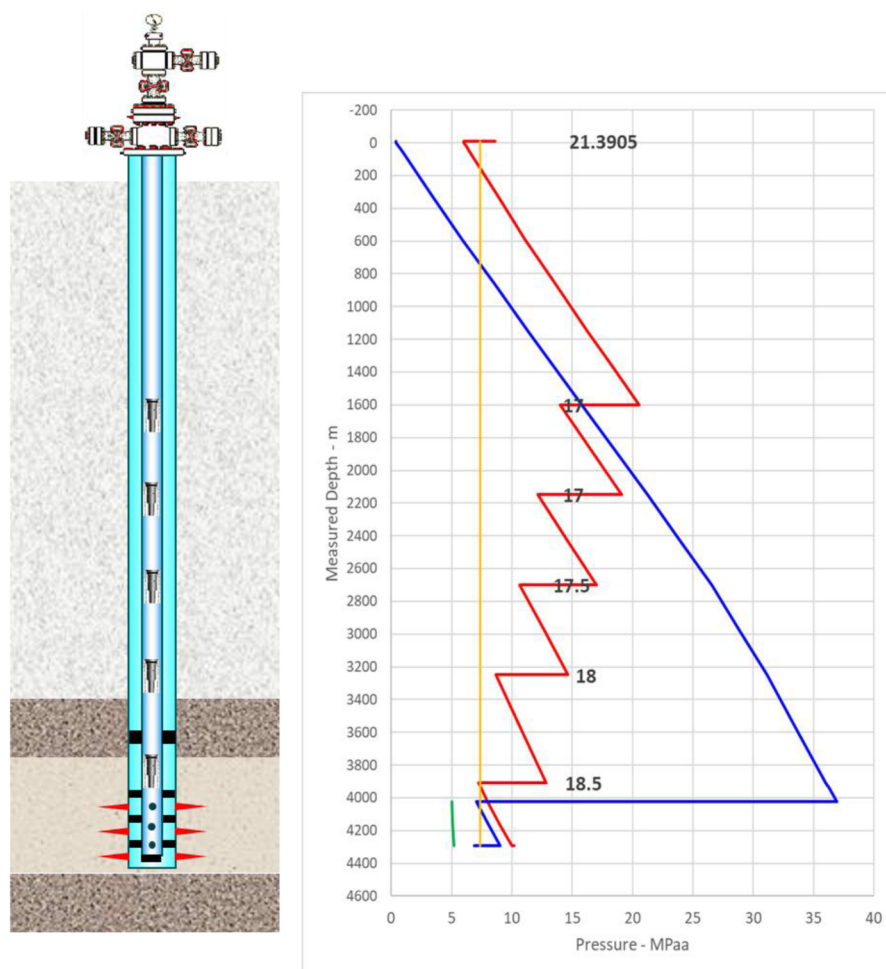


Figure 9—Strategic deployment of ICVs in the wellbore to create a step-wise pressure drop

ICVs have been designed to be wirelessly operated, communicating through electromagnetic transmission, pressure pulses or fluid harmonics. In the CCS injection model, up to five in-line, adjustable flow control devices are strategically positioned along the length of the injection conduit from surface to the reservoir depth conduit to provide a stepwise pressure drop to regulate flow rate and control phase behavior of CO₂.

Modeling the Performance of the CCS Injection Well

A CCS injection model has been developed that simulates the choking effect of downhole flow control devices (FCDs) installed in series at strategically distributed in the completion, sized and numbered to maintain the CO₂ in a dense phase (liquid or supercritical) in the injection tubing. The model can manipulate pressure of the wellhead injection choke and Interval Control Valves (ICVs) to maintain the CO₂ in liquid form throughout the well life and to avoid an excessive pressure drop at any point in the tubing. This is

important to avoid flashing of the CO₂ in the injection tubing that will result in flow instability. The model is capable of representing the thermodynamic properties of CO₂ mixed with contaminants, incorporates heat transfer from the overburden to the wellbore, and treats the flow through chokes, ICV's and represents the pressure drop through passive flow control devices as an isenthalpic thermodynamic process. (Petitt and Konopczynski – 2022) Figure 10 illustrates the pressure and temperature conditions of the CO₂ as it traverses the injection tubing from wellhead to perforations, taking step-wise pressure drops through the recommended ICV completion.

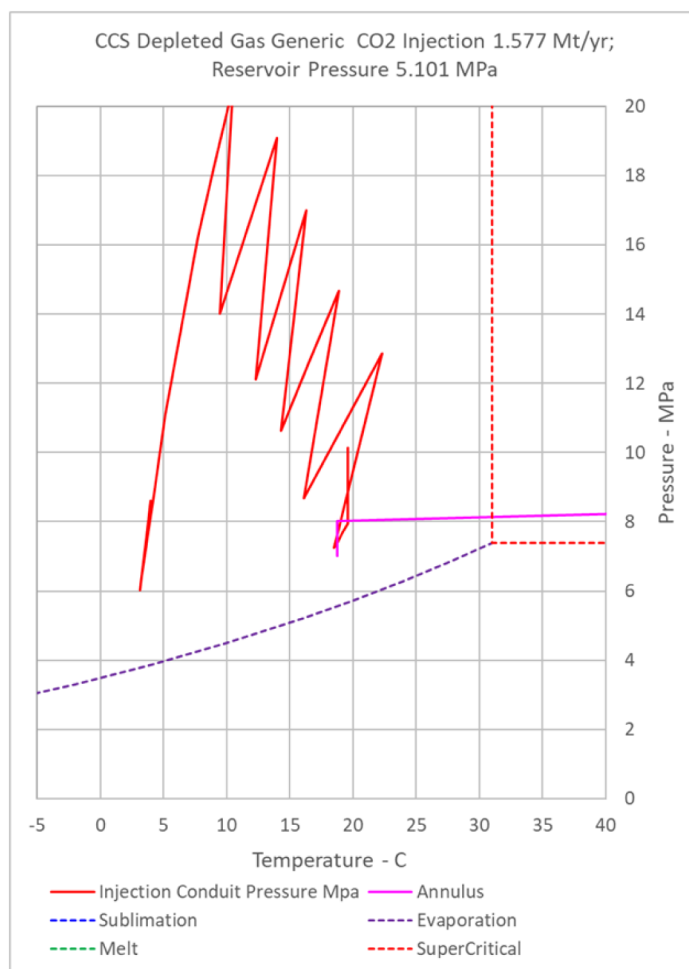


Figure 10—Carbon dioxide pressure-temperature locus within the injection wellbore with recommended ICV completion imposed on the phase diagram for CO₂

Summary

There are many challenges with CO₂ sequestration projects including technical, regulatory and financial hurdles. While most of the technical issues are well understood and there is a foundation of solutions to address the challenges, there is an opportunity to optimise the efficiency and reliability of CCS wells by improving the completion design with flow control devices. The Oil and gas industry is uniquely positioned to bring our technology, experience and project management to provide safe and efficient storage of CO₂ in geologic formations

Acknowledgements

The author wishes to thank TAQA Completions for their permission to publish this paper.

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